2	**D**	intensity I = (Power P) / (surface area A) $I = \dfrac{P}{4\pi r^2}$ $P = 4\pi r^2 I = 4\pi(2.0)^2(0.25) = 12.566\text{ W}$ $\dfrac{\Delta P}{P} = \dfrac{\Delta I}{I} = +2\dfrac{\Delta r}{r}$ $\dfrac{\Delta P}{12.566} = \dfrac{0.05}{0.25} + 2\left(\dfrac{0.1}{2.0}\right) = 0.3$ $\Delta P = 4\text{W}$
3	D	Initially, the speed of the car is increasing in the forward direction. **At X, the car reaches**